

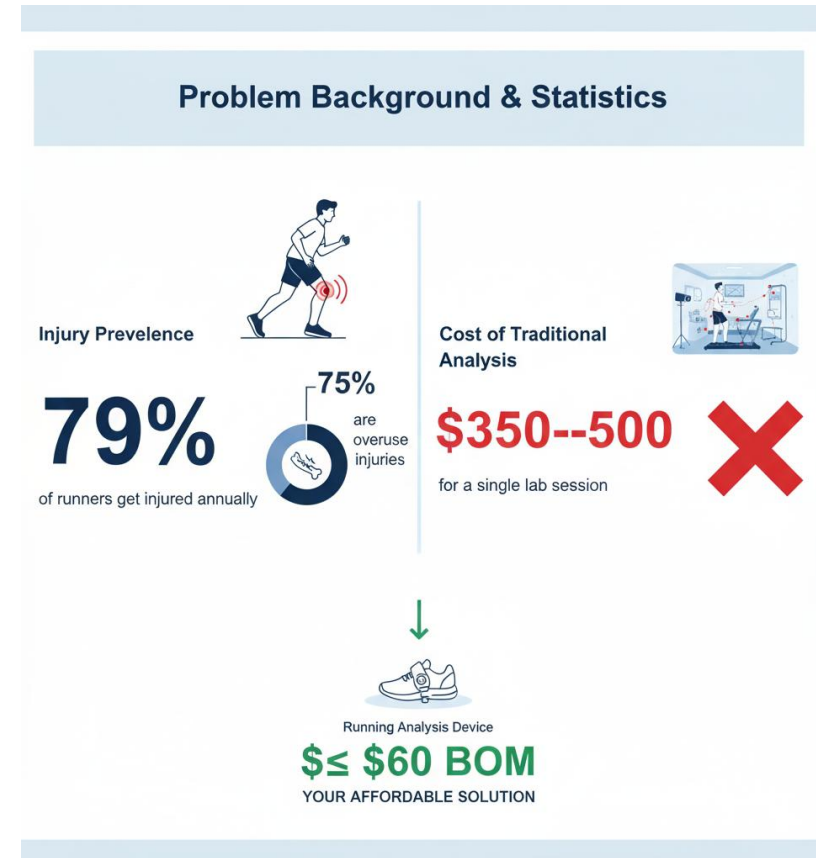
Validation of problem



- **Consumers, Users, and Interviews:** Among high school cross-country runners it was indicated that 60% were unaware of their current running cadence, yet 75% of them expressed concern about past or potential injury. The consensus among users is that professional gait analysis is too expensive and hard to access to be a practical training tool.
- **Scholarly Articles:** Research published in the *Journal of Orthopaedic & Sports Physical Therapy* confirms that increasing a runner's cadence by 5-10% can significantly reduce loading rates at the knee and hip, validating the need for simple cadence feedback.
- **Brief Summary of Existing Products and Patents:** High-end competitive products offer power metrics but are priced at over \$200, confirming the existence of a high-price barrier. Existing patents primarily focus on complex multi-sensor systems, which drives up manufacturing cost.

Statistics:

- Injury Prevalence (Why the Problem Exists)
 - 79%, The estimated amount of runners who sustained an injury that interferes with running activity during any given year.
 - 50-70%, The range of running injuries that are overuse injuries, which are directly linked to repetitive stress from poor biomechanics.
 - 20 to 40 times, The magnitude of increase in the average runner's injury risk when transitioning from low to high mileage without correcting underlying form flaws.
- Biomechanics and Impact (Why the Solution Works)
 - 170-180 steps per minute, The widely accepted "optimal" cadence range for reducing impact forces.
 - 5-10%, The required increase in a runner's cadence shown to significantly reduce stress on the hip, knee, and ankle joints. Your device provides the data needed for this precise adjustment.
 - 2.5-3.5 times body weight, The typical peak vertical ground reaction force experienced during running. Monitoring and reducing force is the best way to prevent bone stress injuries.
- Cost & Accessibility (Why Our Project is Necessary)
 - \$350 to \$500, The average cost for a single, professional in-lab gait analysis session.
 - <\$60, Our project's estimated bill of materials cost.
 - 4 to 8 weeks, The typical length of time a runner may need physical therapy for a moderate overuse injury.

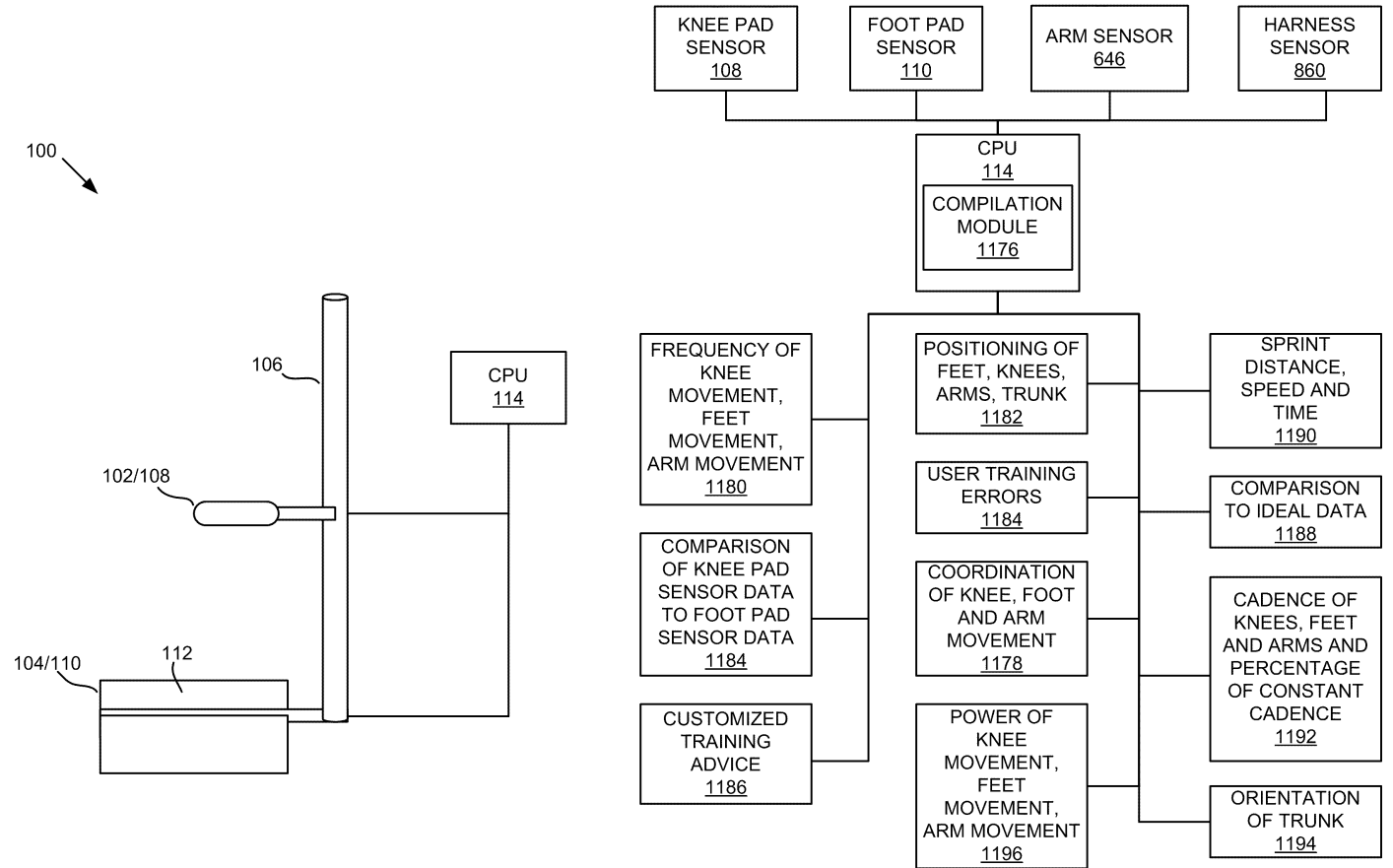


Gathering info

We found our information primarily from online research on various topics within our project. Information we gathered from this research included analyzing both developed commercial products and patented technologies that address gait analysis. One of the main things we were looking at was the cost versus data trade off in existing products that work.



Similar patents



Similar Patents

Abstract

- An apparatus, system, and method are disclosed for accurate measurement, compilation, and presentation of sprinting and running technique data. In one embodiment, the apparatus, system and method includes a knee pad configured to contain a knee pad sensor, a foot pad configured to contain a foot pad sensor and a main trunk that connects the foot pad and the kneepad. The knee pad sensor and the foot pad sensor sense movement near the sensors and contact to the sensors. Data from the sensors is output to a CPU. In one embodiment, the apparatus, system and method include arm sensors. The arm sensors sense arm movement of a user and report arm movement data to the CPU. In a further embodiment, the apparatus may be configured to contain a harness positioned opposite the main trunk.

Date of patent

- 03/08/2011

Patent number

- US7901325B2

Pros

- Provides detailed feedback
- Pinpoints mistakes

Cons

- Clunky
- If the sensors move at all then data become inaccurate

Similar devices



- RunScribe
 - 2 small compact devices that attaches to the laces of your shoes and tracks cadence and stride length and sends that data to an app that they have so that the user can see the data.
- Price
 - 600\$
- Pros
 - Small and compact
 - Measures and returns a bunch of different data
- Cons
 - Expensive
 - Is out of business

Similar devices

- Stryd
 - A compact device that goes on the laces of your shoes and like RunScribe tracks cadence and stride length and sends that data to an app of theirs so that users can see the data
- Price
 - 200\$ for their best version (stryd 5.0)
- Pros
 - Instantaneous feedback
 - Highly accurate
- Cons
 - Expensive
 - Uses a measurement system called power which isn't well known and requires a learning curve





Primary Stake Holder

- **Users:** Recreational runners and high school athletes
- **Buyers:** Parents, coaches, or school athletic departments.
- **Sellers:** E-commerce platforms and/or sporting goods stores.
- **Manufacturers:** Electronics assemblers.

Input on Design Requirements

- **Users/Buyers Input:** We have estimated that cost tolerance would likely be under \$75, and the device must be small enough not to interfere with stride or balance.
- **Manufacturers Input:** Component selection was limited to common, off-the-shelf microcontrollers and sensors (Arduino Nano, MPU-6050) to ensure low BOM and ease of assembly.

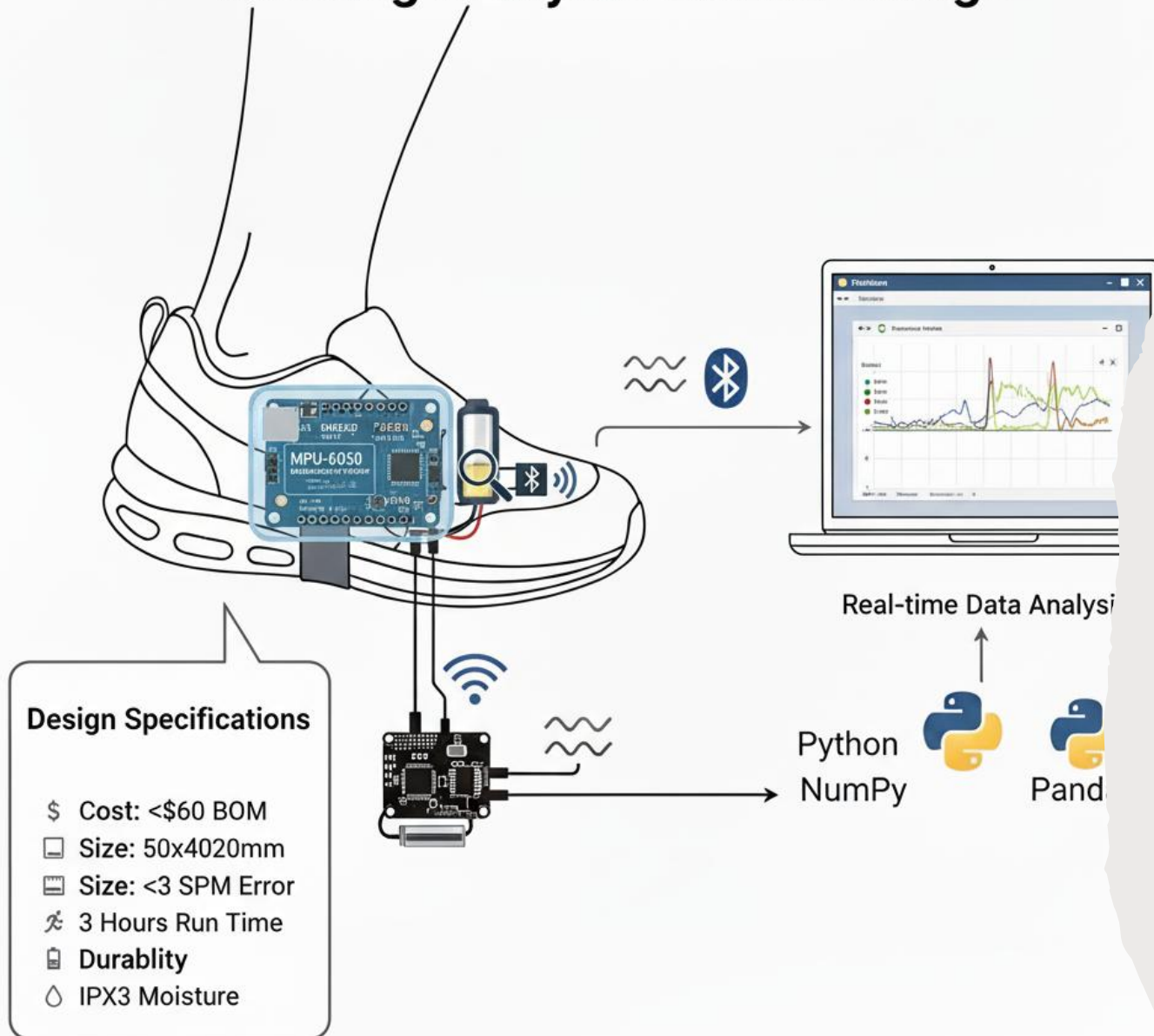
Communication Log

Click here to enter a date.	Correspondence Notes / Email Transcripts
Contact:	Dear <u>Tekscan</u> Performance Team,
Contact Title:	I am an engineering student currently undertaking a research and development project focused on creating an effective, low-cost gait analysis tool. I have familiarized myself with <u>Tekscan's</u> work in force and pressure measurement, and I find your in-shoe systems (like the F-Scan) for dynamic plantar pressure mapping intriguing. I am reaching out for some non-proprietary, general technical insight as I design my own pressure-based system. I understand the thinness and flexibility of your sensing materials are key to integration, and I have specific questions regarding this technology:
Technical Inquiry: Pressure Mapping and Durability in <u>Tekscan</u> In-Shoe Gait Systems	1. Sensor Durability and Material: Given that running involves repetitive, high-impact shear and compression forces, how do you ensure the long-term durability and consistency of the thin sensing material? Are there general insights you can share about the type of flexible sensor array technology best suited for high-cycle, high-impact running analysis?
Contact Information:	2. Calibration and Accuracy: In-shoe pressure sensors are sensitive to temperature, bending, and moisture. What are the general engineering principles used to maintain the calibration and measurement accuracy of the sensors over repeated use, especially when running outdoors?
info@tekscan.com	3. Data Processing (COP/Force): Your software is excellent at generating instantaneous pressure distribution and Center of Pressure data. Can you offer general guidance on the calculation method for generating a highly stable and smooth COP trajectory during the rapid, high-force stance phase of the running gait?
Response:	Any perspective you could offer on these challenges would be immensely helpful to my engineering project. Thank you very much for your time and expertise.
The <u>Tekscan</u> team did not get back to me so I will be sending a follow up email later this week.	v/r Hudson Howland
Follow-up Required - Date	Action Items
9/13/2025	

Communication Log

- I reached out to Tekscan because I like their concept with the shoe insert and the ankle mount.
- Did not receive an email back from their team
- Planning on sending a follow up email next week.

Running Analysis Device Design



Determination of Design Specifications

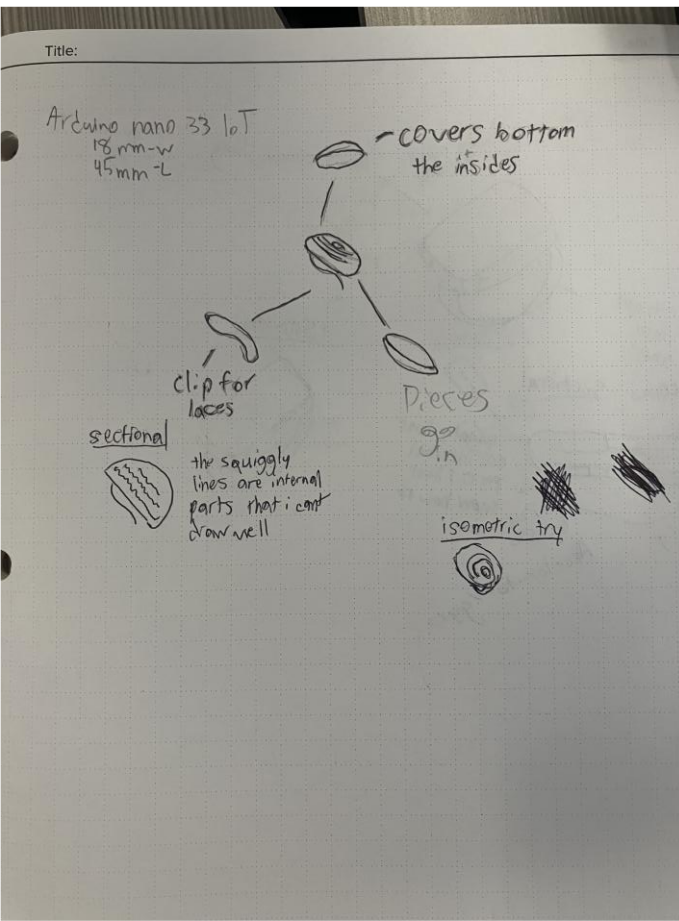
- Our design specifications have been determined by balancing user focused needs with technical feasibility and cost constraints. The core objective is to achieve high accuracy for high-impact events without the high cost associated with lab equipment. Input from target users and technical research guides our prioritization of functional requirements over non-essential features, ensuring a minimum viable product that solves the core problem.

Design Specifications

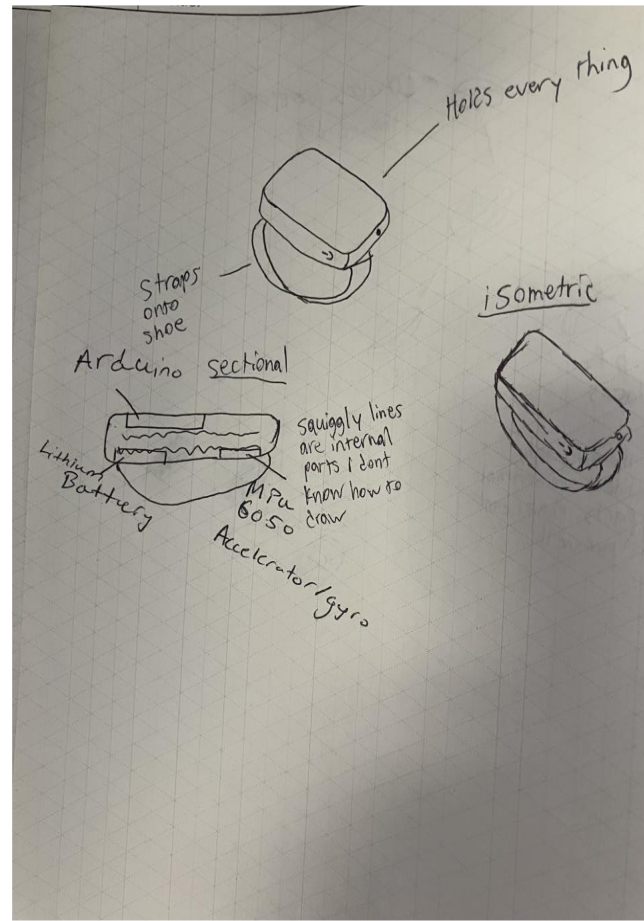
Priority	Specification	Measurable/Clear
1 (Cost)	Total Bill of Materials (BOM) must be less than \$60.00.	Measurable
2 (Functionality)	Must successfully measure and report Cadence within a 3-step margin of error.	Measurable
3 (Portability/Size)	Enclosure maximum dimensions must not exceed 50mm × 40mm × 20mm.	Measurable
4 (Data Rate)	Must maintain a minimum data sampling rate of 100 Hz for accurate peak detection.	Measurable
5 (Battery Life)	Must operate continuously for a minimum of 3 hours on a single charge.	Measurable
6 (Durability)	Device must withstand temperatures from 35 degrees F to 100 degrees F and light moisture (IPX3 equivalent).	Clear

Design concepts

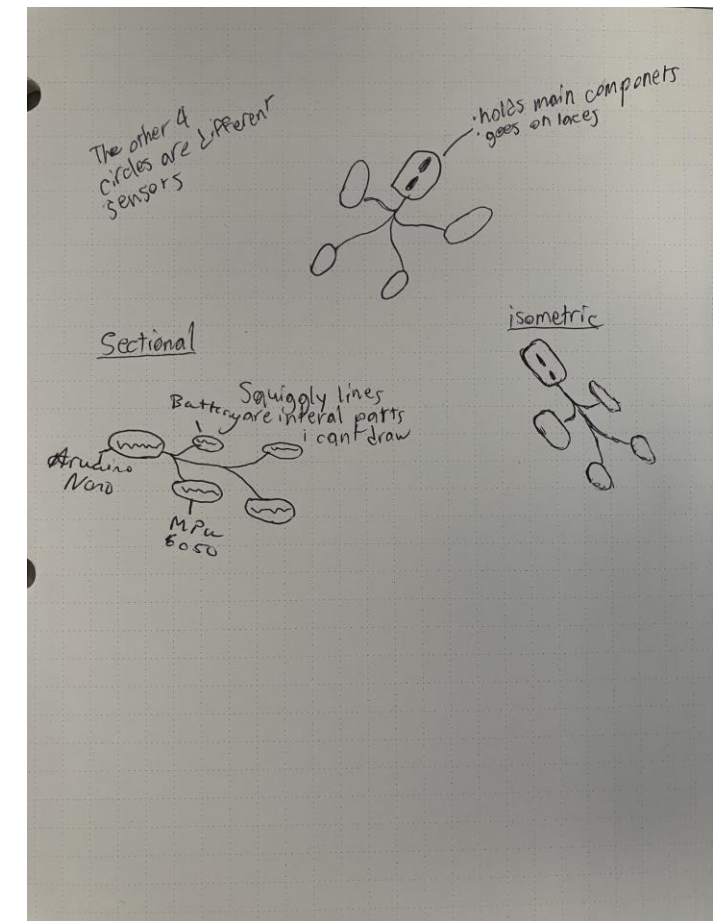
Concept 1



Concept 2



Concept 3



Decision matrix

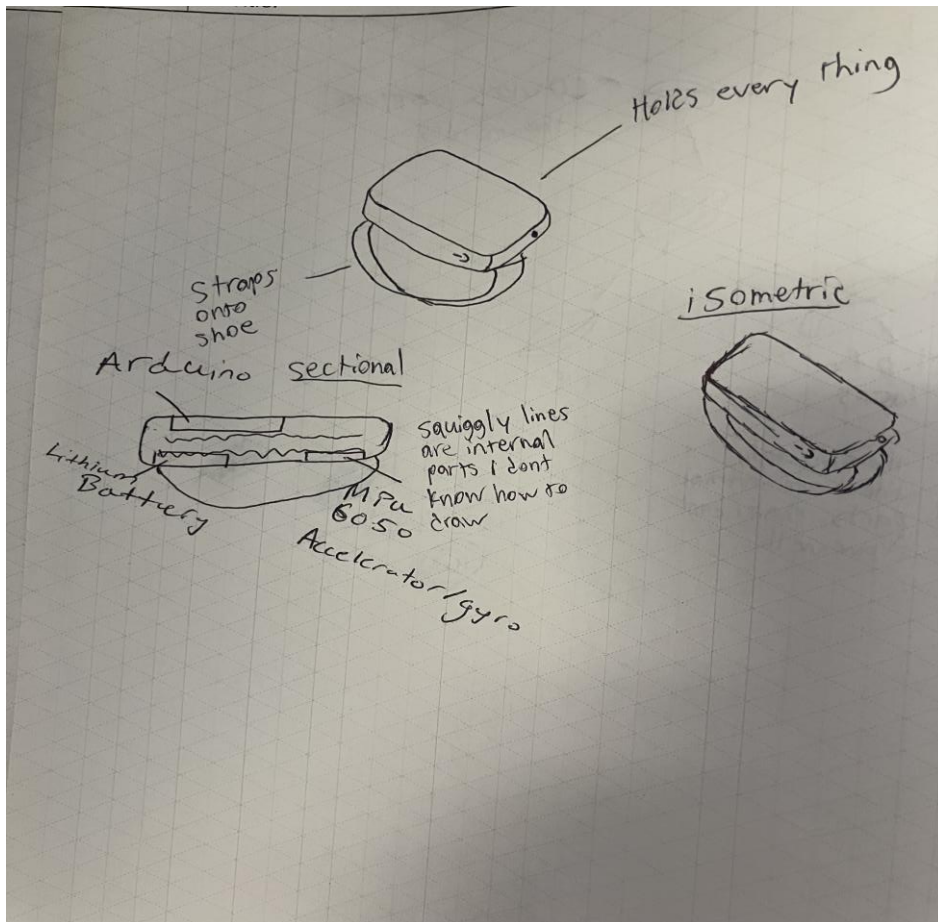
PLTW Engineering

Decision Matrix Template

Ideas	Criteria and Constraints						Totals
	Ease of mass production	durability	creativity	comfortability	Cost effecency	quality	
Idea #1 (our group)	3.5	4.5	3	5	4	4	24
Idea #2 (our group)	5	4.5	2	4	4	5	24.5
Idea #3 (our group)	2	3	5	3	3	3.5	19.5
Idea #1 (Jackson D)	3	4	4	5	4	4	24
Idea #2 (Jackson D)	4	5	3	4	4	5	25
Idea #3 (Jackson D)	2	3	5	5	3	4	22
Idea #1 (Beau)	4	5	3	3	4	5	24
Idea #2 (Beau)	4	4	3	5	5	3	24
Idea #3 (Beau)	2	2	3	1	2	1	11

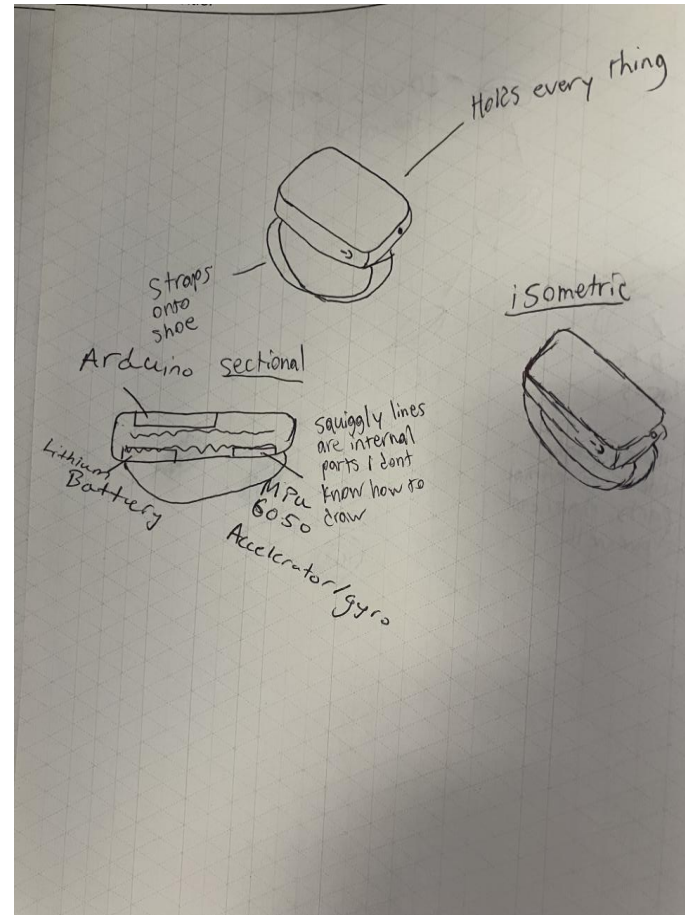
Design concepts: Final Design

Concept 2



- This design had the best total score out of each design we tested with the design matrix.
- Although this design is not the most creative, we determined that it is the best option because it is simple, cost effective, and durable.

Final decision



Final Prototype



Design Features

- **Snap-Fit Assembly:** The box is designed with a two-part construction for easy access to the battery and internal components during the testing phase.
- **Integrated Mounting Points:** The exterior geometry includes specialized slots to accommodate the elastic strap system, ensuring the sensor remains aligned with the foot's longitudinal axis.
- **Internal Layout:** The internal volume is optimized to prevent component shifting, which is critical for maintaining data integrity during high-speed movement.



Manufacturing & Materials

- **Chassis Material:** PLA Filament. We used inventor modeling and 3D printing to create the main enclosure. PLA was selected for its high strength-to-weight ratio and ease of rapid prototyping, ensuring the device meets our Priority constraints.
- **Shock Absorption:** High-Density Rubber Dampening. To mitigate the high G-forces of running impacts, the internal cavity features rubber inserts. These act as a mechanical low-pass filter, protecting the sensitive MPU-6050 sensor and Arduino from high-frequency vibrations and structural shock.

References:

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